CANON AI — Individual Contribution Report

Member: Tanishka Gawale • Role: Principal Project & Reliability Engineer | Team RUNTIME REBELS

Engineering Focus: Campaign Firewall Radar, Merkle Hash Chain Ledger, and QA/Reliability Matrix	**Hackathon Window:** Round 3 Championship (72 Hours)
Live Project: https://canon-ai.vercel.app	**GitHub Repository:** https://github.com/Dhruvplaz-web/Canon-AI
Primary Ownership: Weapon 3 (Sub-12ms Edge Campaign Firewall), Cryptographic Merkle Ledger, and 178 Automated Tests	**Verification Status:** 100% Tested & Merged into `main`

1. Executive Summary of Responsibilities & Impact

In Round 3 of the Codex 48 Hackathon, **Tanishka Gawale** served as **Principal Project & Reliability Engineer** for Team RUNTIME REBELS. Rather than relying on monolithic or single-contributor code, our team distributed platform ownership across specialized engineering, regulatory, design, and reliability domains.

Tanishka Gawale's core responsibility was the end-to-end design, implementation, and rigorous testing of **Weapon 3 (Sub-12ms Edge Campaign Firewall), Cryptographic Merkle Ledger, and 178 Automated Tests**, directly enabling Canon's transformation from a basic text checker into a court-admissible, multi-modal governance powerhouse.

2. Key Features & Concrete Deliverables Built

1. **Weapon 3: Sub-12ms Programmatic Campaign Firewall & Circuit Breaker**: Engineered the automated edge kill switch that intercepts non-compliant live campaigns and issues sub-12ms pause webhooks to Meta Marketing API v19.0 and Google Ads API v16.
2. **Real-Time ■1,200/sec Ad-Spend Pacing Telemetry**: Built live ad-spend burn rate ticker calculating active media spend, budget protected from penalty exposure, and averted CCPA fines in real time.
3. **Cryptographic SHA-256 Merkle Hash Chain Ledger**: Architected an append-only block ledger where every governance action (genesis, submission, check run, override, certificate) is cryptographically linked with SHA-256 hashes.
4. **Interactive Tamper Simulation & Dossier Exporter**: Implemented a 1-click tamper simulation demonstrating Merkle chain breakage, alongside an export engine generating sanitized CSV (with spreadsheet formula injection defense) and JSON audit dossiers.
5. **Full QA Automation & Release Reliability (178/178 Tests)**: Managed the comprehensive testing pipeline across backend Pytest (126 tests) and frontend Jest/MJS (52 tests), achieving 100% pass rates, 0 TypeScript errors, and seamless Vercel deployment.

3. Technical Architecture & Hardest Problem Solved: Sub-12ms Circuit Breaking & Formula-Injection-Proof Dossier Export

The hardest systems challenge was ensuring that the campaign firewall could safely trigger automated circuit breakers without false-positive campaign disruption, while guaranteeing complete data immutability in regulatory exports. Tanishka designed an asynchronous edge webhook dispatcher with optimistic concurrency control. When a campaign's aggregate statutory risk score crosses 85%, the circuit breaker trips within 11.4 milliseconds, dispatching MUTATE requests that halt active ad sets before additional ad spend can be consumed.

In addition, Tanishka engineered the Regulatory Audit Dossier Exporter. To eliminate security vulnerabilities when legal officers open compliance logs in Microsoft Excel, she implemented defensive sanitization against CSV Formula Injection (CWE-1236), prepending single-quote delimiters to any field starting with '=', '+', '-', or '@'. She also established the 178-test automated CI verification pipeline, ensuring zero regressions across all 45 production routes.

DOMAIN VERIFICATION & AUDIT EVIDENCE
Component Owner: Tanishka Gawale (Principal Project & Reliability Engineer)
Module Integrity: Verified across simulated workloads with 0 memory leaks and zero regression failures.
Cryptographic Attestation: SHA-256 block ledger validated against Section 65B Indian Evidence Act.

4. Quality Assurance & Testing Metrics

Verification Target	Test Scope & Method	Result
Circuit Breaker Latency	End-to-end trip time from risk spike to webhook	**11.4ms Average (<12ms Target)**
Merkle Hash Verification	Full chain SHA-256 re-computation and validation	**100% Block Integrity Verified**
CSV Formula Injection	Sanitization against malicious macro/formula cells	**0 Vulnerabilities (CWE-1236)**
Full Test Suite Execution	126 Pytest + 52 Vernacular/Jest tests	**178 / 178 Passed (100%)**

5. Team Peer Review & Verification Attestation

I certify that the architecture, implementation, and test suites described in this document represent the original engineering work of **Tanishka Gawale**, peer-reviewed and verified by all teammates of Team RUNTIME REBELS prior to production release.

Submitted by:	**Verified by Team Leader:**	**Git Release Commit:**
Tanishka Gawale	Dhruv Patil	`ea1014e`
Principal Project & Reliability Engineer	*Team Leader & AI Architect*	*Production Main*